



**WOLLO University**

**Department of software engineering**

**Exit multiple choice questions for**

**Fundamental database system**

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1. What is the full form of DBMS?

- a) Data of Binary Management System
- b) Database Management System
- c) Database Management Service
- d) Data Backup Management System

[View Answer](#)

Answer: b

Explanation: DBMS is abbreviated as Database Management System. Database Management System stores the data and allows authorized users to manipulate and modify the data.

2. What is a database?

- a) Organized collection of information that cannot be accessed, updated, and managed
- b) Collection of data or information without organizing
- c) Organized collection of data or information that can be accessed, updated, and managed
- d) Organized collection of data that cannot be updated

[View Answer](#)

Answer: c

Explanation: It is defined as an organized collection of data or information for easy access, updating, and management in a computer.

3. What is DBMS?

- a) DBMS is a collection of queries
- b) DBMS is a high-level language
- c) DBMS is a programming language
- d) DBMS stores, modifies and retrieves data

[View Answer](#)

Answer: d

Explanation: DBMS is nothing but a storehouse wherein the authorized user can create a database to store, modify or retrieve the organized data in the table. It can be modified or retrieved by users who have access to DBMS only.

4. Who created the first DBMS?

- a) Edgar Frank Codd
- b) Charles Bachman
- c) Charles Babbage
- d) Sharon B. Codd

[View Answer](#)

Answer: b

Explanation: Charles Bachman along with his team invented the first DBMS known as Integrated Data Store (IDS).

5. Which type of data can be stored in the database?

- a) Image oriented data
- b) Text, files containing data
- c) Data in the form of audio or video
- d) All of the above

[View Answer](#)

Answer: d

Explanation: The reason for creating the database management system was to store

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large data and these data can be of any form image, text, audio, or video files, etc. DBMS allows the users to store and access the data of any format.

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6. In which of the following formats data is stored in the database management system?

- a) Image
- b) Text
- c) Table
- d) Graph

[View Answer](#)

Answer: c

Explanation: The data is stored in a table format intended to manage the storage of data and manipulate stored data to generate information.

7. Which of the following is not a type of database?

- a) Hierarchical
- b) Network
- c) Distributed
- d) Decentralized

[View Answer](#)

Answer: d

Explanation: Different types are:

- 1) Centralized
- 2) Distributed
- 3) Relational
- 4) NoSQL
- 5) Cloud
- 6) Object-oriented
- 7) Hierarchical
- 8) Network

8. Which of the following is not an example of DBMS?

- a) MySQL
- b) Microsoft Access
- c) IBM DB2
- d) Google

[View Answer](#)

Answer: d

Explanation: MySQL, Microsoft Access, IBM DB2 are database management systems while Google is a search engine. MySQL is a Linux-based database management system, Microsoft Access is a tool that is a part of Microsoft Office used to store data, IBM DB2 is a database management system developed by IBM. Google's Bigtable is the database that runs Google's Internet search, Google Maps, YouTube, Gmail, and other products.

9. Which of the following is not a feature of DBMS?

- a) Minimum Duplication and Redundancy of Data
- b) High Level of Security
- c) Single-user Access only
- d) Support ACID Property

[View Answer](#)

Answer: c

Explanation: Single-user Access only" is not a feature of DBMS. DBMS allows multiple

users to access and manipulate the database concurrently. It provides mechanisms to handle concurrent access and ensure data consistency and integrity among multiple users.

The important features of a database management system are:

- Minimum Duplication and Redundancy of Data
- High Level of Security
- Multiple-user Access
- Support ACID Property

10. Which of the following is a feature of the database?

- a) No-backup for the data stored
- b) User interface provided
- c) Lack of Authentication
- d) Store data in multiple locations

[View Answer](#)

Answer: b

Explanation: The important features are:

- 1) Provides backup for the data stored by the user and the user can retrieve the data whenever required.
- 2) Provides User-interface to access the data.
- 3) Only authorized users can access the stored data.
- 4) Data is stored in one central location but multiple authorized users can access the data.

11. Which of the following is not a function of the database?

- a) Managing stored data
- b) Manipulating data
- c) Security for stored data
- d) Analysing code

[View Answer](#)

Answer: d

Explanation: It allows authorized users to update, store, manipulate, or access data. Since data is stored in table format it is easy to access the data and perform the required functions. It also removes duplicate and redundant data.

12. Which of the following is a function of the DBMS?

- a) Storing data
- b) Providing multi-users access control
- c) Data Integrity
- d) All of the above

[View Answer](#)

Answer: d

Explanation: The purpose of creating DBMS was to store the data. The data stored in the database management system can be accessed by multiple users if the access is provided. The data stored will be accurate and complete hence providing data integrity.

13. Which of the following is a component of the DBMS?

- a) Data
- b) Data Languages
- c) Data Manager

d) All of the above

[View Answer](#)

Answer: d

Explanation: The components of DBMS are as follows:

- 1) Hardware: Like a hard drive, monitor, etc.
- 2) Software: Provides a user interface
- 3) Data Manager: Manages operations of DBMS.
- 4) Data: The collection of information on the DB is known as data.
- 5) Data Languages: Languages like DDL, DML, DAL, and DCL allow to perform operations like creating, modifying, storing, or retrieving data.

14. Which of the following is known as a set of entities of the same type that share same properties, or attributes?

- a) Relation set
- b) Tuples
- c) Entity set
- d) Entity Relation model

[View Answer](#)

Answer: c

Explanation: In the actual world, an entity is a distinct “thing” or “object” from all other objects. For example: Each employee of an organization is an entity.

15. What is information about data called?

- a) Hyper data
- b) Tera data
- c) Meta data
- d) Relations

[View Answer](#)

Answer: c

Explanation: Information about data is known as Metadata. Metadata describes the data in detail by providing additional information like type, length of the data, etc. Metadata helps the user to understand the data.

16. What does an RDBMS consist of?

- a) Collection of Records
- b) Collection of Keys
- c) Collection of Tables
- d) Collection of Fields

[View Answer](#)

Answer: c

Explanation: It consists of a collection of tables i.e., the data is organized in tabular format. The columns of the relation are known as Fields and rows of the relation are known as fields. Constraints in a relation are known as Keys.

17. The values appearing in given attributes of any tuple in the referencing relation must likewise occur in specified attributes of at least one tuple in the referenced relation, according to \_\_\_\_\_ integrity constraint.

- a) Referential
- b) Primary
- c) Referencing
- d) Specific

[View Answer](#)

Answer: a

Explanation: Consider 2 relations r1 and r2. r1 may include among its attributes the primary key of relation r2. This attribute is called a foreign key from r1, referencing r2. The relation r1 is also called the referencing relation of the foreign key dependency, and r2 is called the referenced relation of the foreign key.

18. \_\_\_\_\_ is a hardware component that is most important for the operation of a database management system.

- a) Microphone
- b) High speed, large capacity disk to store data
- c) High-resolution video display
- d) Printer

[View Answer](#)

Answer: b

Explanation: Since all the data are stored in form of memory in the disk, a high speed, and large-capacity disk is required for the operation of the database management system.

19. The DBMS acts as an interface between \_\_\_\_\_ and \_\_\_\_\_ of an enterprise-class system.

- a) Data and the DBMS
- b) Application and SQL
- c) Database application and the database
- d) The user and the software

[View Answer](#)

Answer: c

Explanation: A database management system is an application that acts as an interface between the user and the database. The user interacts with the application to access data.

20. The ability to query data, as well as insert, delete, and alter tuples, is offered by \_\_\_\_\_

- a) TCL (Transaction Control Language)
- b) DCL (Data Control Language)
- c) DDL (Data Definition Language)
- d) DML (Data Manipulation Language)

[View Answer](#)

Answer: d

Explanation: A query is a request for data or information. Relational Schema is the design and structure of the relation. DDL consists of commands that help in modifying. DML performs the change in the values of the relation.

21. \_\_\_\_\_ is a set of one or more attributes taken collectively to uniquely identify a record.

- a) Primary Key
- b) Foreign key
- c) Super key
- d) Candidate key

[View Answer](#)

Answer: c

Explanation: Foreign key creates a relationship between two relations. Super key is the

superset of all the keys in a relation. A candidate key is used to identify tuples in a relation.

22. Which command is used to remove a relation from an SQL?

- a) Drop table
- b) Delete
- c) Purge
- d) Remove

[View Answer](#)

Answer: a

Explanation: Delete command is used to delete the existing record from the table. The drop table deletes the whole structure of the relation. Purge removes the table which cannot be obtained again.

23. Which of the following set should be associated with weak entity set for weak entity to be meaningful?

- a) Neighbour set
- b) Strong entity set
- c) Owner set
- d) Identifying set

[View Answer](#)

Answer: d

Explanation: Every weak entity must be linked to an identifying entity; in other words, the existence of the weak entity set is contingent on the presence of the identifying entity set. The weak entity set that the identifying entity set identifies is said to be owned by the identifying entity set. Owner entity set is another name for it.

24. Which of the following command is correct to delete the values in the relation teaches?

- a) Delete from teaches;
- b) Delete from teaches where Id = 'Null';
- c) Remove table teaches;
- d) Drop table teaches;

[View Answer](#)

Answer: a

Explanation: To delete the entries from the table Delete from table command should be used.

25. Procedural language among the following is \_\_\_\_\_

- a) Domain relational calculus
- b) Tuple relational calculus
- c) Relational algebra
- d) Query language

[View Answer](#)

Answer: c

Explanation: Non-Procedural Languages are Domain relational calculus and Tuple relational calculus. Relational algebra is a procedural language that takes input in the form of relation and output generated is also a relation.

26. \_\_\_\_\_ operations do not preserve non-matched tuples.

- a) Left outer join
- b) Inner join
- c) Natural join

d) Right outer join

[View Answer](#)

Answer: b

Explanation: Left outer join returns all the rows from the table that is on the left side and matching rows on the right side of the join. Inner join returns all rows when there is at least one match in BOTH tables. Natural join returns the common columns from the tables being joined. A right outer join returns all the rows from the table that is on the right side and matching rows on the left side of the join.

27. Which forms have a relation that contains information about a single entity?

a) 4NF

b) 2NF

c) 5NF

d) 3NF

[View Answer](#)

Answer: a

Explanation: If and only if, for each of its non-trivial multivalued dependencies  $X \twoheadrightarrow Y$ , a table is in 4NF. X is a superkey—that is, X is either a candidate key or a superset thereof.

28. The top level of the hierarchy consists of \_\_\_\_\_ each of which can contain \_\_\_\_\_.

a) Schemas, Catalogs

b) Schemas, Environment

c) Environment, Schemas

d) Catalogs, Schemas

[View Answer](#)

Answer: d

Explanation: Schemas represent the logical configuration of the DBMS. Catalogs consist of metadata of the objects and system settings used.

29. \_\_\_\_\_ indicates the maximum number of entities that can be involved in a relationship.

a) Greater Entity Count

b) Minimum cardinality

c) Maximum cardinality

d) ERD

[View Answer](#)

Answer: c

Explanation: The term cardinality refers to the uniqueness of data values included in a single column (attribute) of a table in SQL (Structured Query Language).

30. The user IDs can be added or removed using which of the following fixed roles?

a) db\_sysadmin

b) db\_accessadmin

c) db\_securityadmin

d) db\_setupadmin

[View Answer](#)

Answer: b

Explanation: Along with security, as the name suggests the db\_accessadmin role also handles access. db\_sysadmin refers to the system administrator. db\_securityadmin as the name suggests it involves granting or declining permission to access the data ensuring security.

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31. Why the following statement is erroneous?

```
SELECT dept_name, ID, avg (salary)
FROM instructor
GROUP BY dept_name;
```

- a) Dept\_id should not be used in group by clause
- b) Group by clause is not valid in this query
- c) Avg(salary) should not be selected
- d) None

[View Answer](#)

Answer: a

Explanation: Any property that does not occur in the group by clause must only appear in an aggregate function if it also appears in the select clause; otherwise, the query is considered incorrect.

32. The traditional storage of data organized by the customer, stored in separate folders in filing cabinets is an example of \_\_\_\_\_ type of 'database' management system.

- a) Object-oriented database management system
- b) Relational database management system
- c) Network database management system
- d) Hierarchical database management system

[View Answer](#)

Answer: d

Explanation: In an object-oriented database management system, the data is stored in the form of objects. In a relational DBMS, the data is stored in the form of tables.

Hierarchy is obtained by Parent-Child Relationship. Parent-Child Relationship Type is basically a 1:N relationship.

33. After groups have been established, SQL applies predicates in the \_\_\_\_\_ clause, allowing aggregate functions to be used.

- a) Where
- b) Having
- c) Group by
- d) With

[View Answer](#)

Answer: d

Explanation: The "with" clause was introduced to reduce the complexity of the queries which are lengthy. The "with" clause allows you to define a temporary relation whose definition is only visible to the query in which it appears.

34. Which of the following is not the utility of DBMS?

- i) Backup ii) Loading iii) Process Organization iv) File organization
- a) i, ii, and iv only
- b) i, ii and iii only
- c) i, iii and iv only
- d) All i, ii, iii, and iv

[View Answer](#)

Answer: a

Explanation: Backup utility is used to create a copy of the db as a backup. Loading utility is used to load existing file. File organization is used to relocate the files and create new access path. Processing is not an utility.

35. What does a foreign key combined with a primary key create?

- a) Network model between the tables that connect them
- b) Parent-Child relationship between the tables that connects them
- c) One to many relationship between the tables that connects them
- d) All of the mentioned

[View Answer](#)

Answer: a

Explanation: Using the two relationships mother and father gives us a record of a child's mother, even if we don't know who the father is; if the ternary connection parent is used, a null value is necessary. In this scenario, binary relationship sets are preferred.

36. Which of the following is correct according to the technology deployed by DBMS?

- a) Pointers are used to maintain transactional integrity and consistency
- b) Cursors are used to maintain transactional integrity and consistency
- c) Locks are used to maintain transactional integrity and consistency
- d) Triggers are used to maintain transactional integrity and consistency

[View Answer](#)

Answer: c

Explanation: Pointers are used to access data with great speed and accuracy. Consistency is maintained using locks.

37. Which of the following is correct regarding the file produced by a spreadsheet?

- a) can be used as it is by the DBMS
- b) stored on disk in an ASCII text format
- c) all of the mentioned
- d) none of the mentioned

[View Answer](#)

Answer: a

Explanation: For updating the value in ASCII text format, a regular text file is used.

38. What is the function of the following command?

Delete from r where P;

- a) Clears entries from relation
- b) Deletes relation
- c) Deletes particular tuple from relation
- d) All of the mentioned

[View Answer](#)

Answer: c

Explanation: In the command the P gives condition to delete a particular tuple.

39. \_\_\_\_\_ resembles Create view.

- a) Create table . . . as
- b) Create view as
- c) Create table . . .like
- d) With data

[View Answer](#)

Answer: a

Explanation: The 'create table... as' statement is similar to the 'create view... as' statement in that both are defined with queries. The main distinction is that table

contents are set when the table is built, whereas view contents always reflect the current query result.

40. The query specifying the SQL view is said to be updatable if it meets which of the following conditions?

- a) select clause contains relation attribute names but not have expressions, aggregates, or distinct specification
- b) from clause has 1 relation
- c) query does not have group by or having clause
- d) All of the mentioned

[View Answer](#)

Answer: d

Explanation: To update the view in sql all the conditions must be satisfied.

41. When the “ROLLUP” operator for expression or columns within a “GROUP BY” clause is used?

- a) Find the groups that make up the subtotal in a row
- b) Create group-wise grand totals for the groups indicated in a GROUP BY clause
- c) Group expressions or columns specified in a GROUP BY clause in one direction, from right to left, for computing the subtotals
- d) To produce a cross-tabular report for computing subtotals by grouping phrases or columns given within a GROUP BY clause in all available directions

[View Answer](#)

Answer: c

Explanation: A view can be thought of as a virtual table that gets its data from one or more table columns.

42. Which of the following is the best way to represent the attributes in a large db?

- a) Dot representation
- b) Concatenation
- c) Relational-and
- d) All of the mentioned

[View Answer](#)

Answer: b

Explanation: Concatenation in DBMS is used to join two or more table fields of the same table or different tables. Example inst sec and student sec.

43. Which of the following is the subset of SQL commands used to manipulate Oracle Structures, including tables?

- a) Data Described Language
- b) Data Retrieval Language
- c) Data Manipulation Language
- d) Data Definition Language

[View Answer](#)

Answer: d

Explanation: DDLs are used to define schema and table characters and consist of commands that help in modifying. DML performs the change in the values of the relation.

44. Which of the following functions construct histograms and use buckets for ranking?

- a) Ntil()
- b) Newtil()
- c) Rank()

d) All of the mentioned

[View Answer](#)

Answer: a

Explanation: ntile(n) returns the number of the bucket in which each tuple is stored, with bucket numbers beginning with 1.

45. \_\_\_\_\_ command is used in SQL to issue multiple CREATE TABLE, CREATE VIEW and GRANT statements in a single transaction.

- a) CREATE CLUSTER
- b) CREATE PACKAGE
- c) CREATE SCHEMA
- d) All of the mentioned

[View Answer](#)

Answer: c

Explanation: A schema is a description of a system's structure in a formal language supported by the database management system, and it refers to data organization as a blueprint for how a db is built.

46. Which of the following key is required in to handle the data when the encryption is applied to the data so that the unauthorised user cannot access the data?

- a) Primary key
- b) Authorised key
- c) Encryption key
- d) Decryption key

[View Answer](#)

Answer: d

Explanation: The encryption key is used to encrypt the message. Even if the message is intercepted by an enemy, the enemy will be unable to decrypt and interpret the message because he lacks the key.

47. Which of the following is known as the process of viewing cross-tab with a fixed value of one attribute?

- a) Dicing
- b) Pivoting
- c) Slicing
- d) Both Pivoting and Dicing

[View Answer](#)

Answer: c

Explanation: Slice procedure takes one dimension from a cube and turns it into a new sub-cube. Dice takes two or more dimensions from a cube and creates a new sub-cube from them.

48. For designing a normal RDBMS which of the following normal form is considered adequate?

- a) 4NF
- b) 3NF
- c) 2NF
- d) 5NF

[View Answer](#)

Answer: b

Explanation: Because most 3NF tables are free of insertion, update, and deletion

anomalies, an RDBMS table is sometimes regarded as “normalized” if it is in the Third Normal Form.

49. Which of the following is popular for applications such as storage of log files in a database management system since it offers the best write performance?

- a) RAID level 0
- b) RAID level 1
- c) RAID level 2
- d) RAID level 3

[View Answer](#)

Answer: b

Explanation: RAID level 0 refers to data striping. RAID level 1 refers to disk mirroring with block striping. RAID level 2 refers to bit-level striping and RAID level 3 refers to byte-level striping with dedicated parity.

50. Which of the following represents a query in the tuple relational calculus?

- a)  $\{ \{P(t) \mid t\} \}$
- b)  $\{t \mid P(t)\}$
- c)  $t \mid P() \mid t$
- d) All of the mentioned

[View Answer](#)

Answer: b

Explanation: A nonprocedural query language is the tuple relational calculus. It specifies the needed information but does not provide a detailed strategy for obtaining it.

51. The oldest DB model is \_\_\_\_\_

- a) Network
- b) Physical
- c) Hierarchical
- d) Relational

[View Answer](#)

Answer: a

Explanation: Network model has data stored in a hierarchical network flow. In a relational DBMS, the data is stored in the form of tables. Hierarchy is obtained by Parent-Child Relationship

52. Evaluate the statements issued by the DBA in the given sequence if OE and SCOTT are the users and the ORDERS table is owned by OE.

```
CREATE ROLE r1;  
GRANT SELECT, INSERT ON oe.orders TO r1;  
GRANT r1 TO scott;  
GRANT SELECT ON oe.orders TO scott;  
REVOKE SELECT ON oe.orders FROM scott;
```

What would be the outcome after executing the statements?

- a) The REVOKE statement would give an error because the SELECT privilege has been granted to the role R1
- b) The REVOKE statement would remove the SELECT privilege from SCOTT as well as from the role R1
- c) SCOTT would be able to query the OE.ORDERS table
- d) SCOTT would not be able to query the OE.ORDERS table

[View Answer](#)

Answer: c

Explanation: To perform operations on objects, the REVOKE statement is used to revoke rights from a single user or role, or from all users.

53. Which of the following establishes a top-to-bottom relationship among the items?

- a) Relational schema
- b) Network schema
- c) Hierarchical schema
- d) All of the mentioned

[View Answer](#)

Answer: c

Explanation: A data model in which the data is structured into a tree-like structure is known as a hierarchical model. The structure allows information to be represented using parent-child relationships.

54. A major goal of the db system is to minimize the number of block transfers between the disk and memory. Which of the following helps in achieving this goal?

- a) Secondary storage
- b) Storage
- c) Catalog
- d) Buffer

[View Answer](#)

Answer: d

Explanation: Every block has a copy on disc, however, the copy on the disc may be an earlier version of the block than the version in the buffer.

55. What happens if a piece of data is stored in two places in the db?

- a) Storage space is wasted & Changing the data in one spot will cause data inconsistency
- b) It can be more easily accessed
- c) Changing the data in one spot will cause data inconsistency
- d) Storage space is wasted

[View Answer](#)

Answer: a

Explanation: One of the main features of a database management system is minimum data duplication and redundancy. Hence, it is always consistent and so there is no duplication.

56. The logical design, and the snapshot of the data at a given instant in time is known as?

- a) Instance & Relation
- b) Relation & Schema
- c) Domain & Schema
- d) Schema & Instance

[View Answer](#)

Answer: d

Explanation: Instance is an instance of time, the relation is also known as table consists of data with similar characteristics, Domain is the collection of values that an attribute can contain and schema is a representation.

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- b) Database Management System
- c) Database Management Service
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[View Answer](#)

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Explanation: It is defined as an organized collection of data or information for easy access, updating, and management in a computer.

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[View Answer](#)

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Explanation: Charles Bachman along with his team invented the first DBMS known as Integrated Data Store (IDS).

5. Which type of data can be stored in the database?

- a) Image oriented data
- b) Text, files containing data
- c) Data in the form of audio or video
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[View Answer](#)

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Explanation: The reason for creating the database management system was to store large data and these data can be of any form image, text, audio, or video files, etc. DBMS allows the users to store and access the data of any format.

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6. In which of the following formats data is stored in the database management system?

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[View Answer](#)

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[View Answer](#)

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Explanation: Different types are:

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- 5) Cloud
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- 7) Hierarchical
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- b) Microsoft Access
- c) IBM DB2
- d) Google

[View Answer](#)

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Explanation: MySQL, Microsoft Access, IBM DB2 are database management systems while Google is a search engine. MySQL is a Linux-based database management system, Microsoft Access is a tool that is a part of Microsoft Office used to store data, IBM DB2 is a database management system developed by IBM. Google's Bigtable is the database that runs Google's Internet search, Google Maps, YouTube, Gmail, and other products.

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[View Answer](#)

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- c) Lack of Authentication
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[View Answer](#)

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Explanation: The important features are:

- 1) Provides backup for the data stored by the user and the user can retrieve the data whenever required.
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- 3) Only authorized users can access the stored data.
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[View Answer](#)

Answer: d

Explanation: It allows authorized users to update, store, manipulate, or access data. Since data is stored in table format it is easy to access the data and perform the required functions. It also removes duplicate and redundant data.

12. Which of the following is a function of the DBMS?

- a) Storing data
- b) Providing multi-users access control
- c) Data Integrity
- d) All of the above

[View Answer](#)

Answer: d

Explanation: The purpose of creating DBMS was to store the data. The data stored in the database management system can be accessed by multiple users if the access is provided. The data stored will be accurate and complete hence providing data integrity.

13. Which of the following is a component of the DBMS?

- a) Data

- b) Data Languages
- c) Data Manager
- d) All of the above

[View Answer](#)

Answer: d

Explanation: The components of DBMS are as follows:

- 1) Hardware: Like a hard drive, monitor, etc.
- 2) Software: Provides a user interface
- 3) Data Manager: Manages operations of DBMS.
- 4) Data: The collection of information on the DB is known as data.
- 5) Data Languages: Languages like DDL, DML, DAL, and DCL allow to perform operations like creating, modifying, storing, or retrieving data.

14. Which of the following is known as a set of entities of the same type that share same properties, or attributes?

- a) Relation set
- b) Tuples
- c) Entity set
- d) Entity Relation model

[View Answer](#)

Answer: c

Explanation: In the actual world, an entity is a distinct “thing” or “object” from all other objects. For example: Each employee of an organization is an entity.

15. What is information about data called?

- a) Hyper data
- b) Tera data
- c) Meta data
- d) Relations

[View Answer](#)

Answer: c

Explanation: Information about data is known as Metadata. Metadata describes the data in detail by providing additional information like type, length of the data, etc. Metadata helps the user to understand the data.

16. What does an RDBMS consist of?

- a) Collection of Records
- b) Collection of Keys
- c) Collection of Tables
- d) Collection of Fields

[View Answer](#)

Answer: c

Explanation: It consists of a collection of tables i.e., the data is organized in tabular format. The columns of the relation are known as Fields and rows of the relation are known as fields. Constraints in a relation are known as Keys.

17. The values appearing in given attributes of any tuple in the referencing relation must likewise occur in specified attributes of at least one tuple in the referenced relation, according to \_\_\_\_\_ integrity constraint.

- a) Referential
- b) Primary
- c) Referencing

d) Specific

[View Answer](#)

Answer: a

Explanation: Consider 2 relations r1 and r2. r1 may include among its attributes the primary key of relation r2. This attribute is called a foreign key from r1, referencing r2. The relation r1 is also called the referencing relation of the foreign key dependency, and r2 is called the referenced relation of the foreign key.

18. \_\_\_\_\_ is a hardware component that is most important for the operation of a database management system.

- a) Microphone
- b) High speed, large capacity disk to store data
- c) High-resolution video display
- d) Printer

[View Answer](#)

Answer: b

Explanation: Since all the data are stored in form of memory in the disk, a high speed, and large-capacity disk is required for the operation of the database management system.

19. The DBMS acts as an interface between \_\_\_\_\_ and \_\_\_\_\_ of an enterprise-class system.

- a) Data and the DBMS
- b) Application and SQL
- c) Database application and the database
- d) The user and the software

[View Answer](#)

Answer: c

Explanation: A database management system is an application that acts as an interface between the user and the database. The user interacts with the application to access data.

20. The ability to query data, as well as insert, delete, and alter tuples, is offered by \_\_\_\_\_

- a) TCL (Transaction Control Language)
- b) DCL (Data Control Language)
- c) DDL (Data Definition Language)
- d) DML (Data Manipulation Language)

[View Answer](#)

Answer: d

Explanation: A query is a request for data or information. Relational Schema is the design and structure of the relation. DDL consists of commands that help in modifying. DML performs the change in the values of the relation.

21. \_\_\_\_\_ is a set of one or more attributes taken collectively to uniquely identify a record.

- a) Primary Key
- b) Foreign key
- c) Super key
- d) Candidate key

[View Answer](#)

Answer: c

Explanation: Foreign key creates a relationship between two relations. Super key is the superset of all the keys in a relation. A candidate key is used to identify tuples in a relation.

22. Which command is used to remove a relation from an SQL?

- a) Drop table
- b) Delete
- c) Purge
- d) Remove

[View Answer](#)

Answer: a

Explanation: Delete command is used to delete the existing record from the table. The drop table deletes the whole structure of the relation. Purge removes the table which cannot be obtained again.

23. Which of the following set should be associated with weak entity set for weak entity to be meaningful?

- a) Neighbour set
- b) Strong entity set
- c) Owner set
- d) Identifying set

[View Answer](#)

Answer: d

Explanation: Every weak entity must be linked to an identifying entity; in other words, the existence of the weak entity set is contingent on the presence of the identifying entity set. The weak entity set that the identifying entity set identifies is said to be owned by the identifying entity set. Owner entity set is another name for it.

24. Which of the following command is correct to delete the values in the relation teaches?

- a) Delete from teaches;
- b) Delete from teaches where Id ='Null';
- c) Remove table teaches;
- d) Drop table teaches;

[View Answer](#)

Answer: a

Explanation: To delete the entries from the table Delete from table command should be used.

25. Procedural language among the following is \_\_\_\_\_

- a) Domain relational calculus
- b) Tuple relational calculus
- c) Relational algebra
- d) Query language

[View Answer](#)

Answer: c

Explanation: Non-Procedural Languages are Domain relational calculus and Tuple relational calculus. Relational algebra is a procedural language that takes input in the form of relation and output generated is also a relation.

26. \_\_\_\_\_ operations do not preserve non-matched tuples.

- a) Left outer join

- b) Inner join
- c) Natural join
- d) Right outer join

[View Answer](#)

Answer: b

Explanation: Left outer join returns all the rows from the table that is on the left side and matching rows on the right side of the join. Inner join returns all rows when there is at least one match in BOTH tables. Natural join returns the common columns from the tables being joined. A right outer join returns all the rows from the table that is on the right side and matching rows on the left side of the join.

27. Which forms have a relation that contains information about a single entity?

- a) 4NF
- b) 2NF
- c) 5NF
- d) 3NF

[View Answer](#)

Answer: a

Explanation: If and only if, for each of its non-trivial multivalued dependencies  $X \twoheadrightarrow Y$ , a table is in 4NF.  $X$  is a superkey—that is,  $X$  is either a candidate key or a superset thereof.

28. The top level of the hierarchy consists of \_\_\_\_\_ each of which can contain \_\_\_\_\_.

- a) Schemas, Catalogs
- b) Schemas, Environment
- c) Environment, Schemas
- d) Catalogs, Schemas

[View Answer](#)

Answer: d

Explanation: Schemas represent the logical configuration of the DBMS. Catalogs consist of metadata of the objects and system settings used.

29. \_\_\_\_\_ indicates the maximum number of entities that can be involved in a relationship.

- a) Greater Entity Count
- b) Minimum cardinality
- c) Maximum cardinality
- d) ERD

[View Answer](#)

Answer: c

Explanation: The term cardinality refers to the uniqueness of data values included in a single column (attribute) of a table in SQL (Structured Query Language).

30. The user IDs can be added or removed using which of the following fixed roles?

- a) db\_sysadmin
- b) db\_accessadmin
- c) db\_securityadmin
- d) db\_setupadmin

[View Answer](#)

Answer: b

Explanation: Along with security, as the name suggests the db\_accessadmin role also handles access. db\_sysadmin refers to the system administrator. db\_securityadmin as

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the name suggests it involves granting or declining permission to access the data ensuring security.

31. Why the following statement is erroneous?

```
SELECT dept_name, ID, avg (salary)
FROM instructor
GROUP BY dept_name;
```

- a) Dept\_id should not be used in group by clause
- b) Group by clause is not valid in this query
- c) Avg(salary) should not be selected
- d) None

[View Answer](#)

Answer: a

Explanation: Any property that does not occur in the group by clause must only appear in an aggregate function if it also appears in the select clause; otherwise, the query is considered incorrect.

32. The traditional storage of data organized by the customer, stored in separate folders in filing cabinets is an example of \_\_\_\_\_ type of 'database' management system.

- a) Object-oriented database management system
- b) Relational database management system
- c) Network database management system
- d) Hierarchical database management system

[View Answer](#)

Answer: d

Explanation: In an object-oriented database management system, the data is stored in the form of objects. In a relational DBMS, the data is stored in the form of tables. Hierarchy is obtained by Parent-Child Relationship. Parent-Child Relationship Type is basically a 1:N relationship.

33. After groups have been established, SQL applies predicates in the \_\_\_\_\_ clause, allowing aggregate functions to be used.

- a) Where
- b) Having
- c) Group by
- d) With

[View Answer](#)

Answer: d

Explanation: The "with" clause was introduced to reduce the complexity of the queries which are lengthy. The "with" clause allows you to define a temporary relation whose definition is only visible to the query in which it appears.

34. Which of the following is not the utility of DBMS?

- i) Backup ii) Loading iii) Process Organization iv) File organization
- a) i, ii, and iv only
- b) i, ii and iii only
- c) i, iii and iv only
- d) All i, ii, iii, and iv

[View Answer](#)

Answer: a

Explanation: Backup utility is used to create a copy of the db as a backup. Loading utility is used to load existing file. File organization is used to relocate the files and create new access path. Processing is not an utility.

35. What does a foreign key combined with a primary key create?

- a) Network model between the tables that connect them
- b) Parent-Child relationship between the tables that connects them
- c) One to many relationship between the tables that connects them
- d) All of the mentioned

[View Answer](#)

Answer: a

Explanation: Using the two relationships mother and father gives us a record of a child's mother, even if we don't know who the father is; if the ternary connection parent is used, a null value is necessary. In this scenario, binary relationship sets are preferred.

36. Which of the following is correct according to the technology deployed by DBMS?

- a) Pointers are used to maintain transactional integrity and consistency
- b) Cursors are used to maintain transactional integrity and consistency
- c) Locks are used to maintain transactional integrity and consistency
- d) Triggers are used to maintain transactional integrity and consistency

[View Answer](#)

Answer: c

Explanation: Pointers are used to access data with great speed and accuracy. Consistency is maintained using locks.

37. Which of the following is correct regarding the file produced by a spreadsheet?

- a) can be used as it is by the DBMS
- b) stored on disk in an ASCII text format
- c) all of the mentioned
- d) none of the mentioned

[View Answer](#)

Answer: a

Explanation: For updating the value in ASCII text format, a regular text file is used.

38. What is the function of the following command?

Delete from r where P;

- a) Clears entries from relation
- b) Deletes relation
- c) Deletes particular tuple from relation
- d) All of the mentioned

[View Answer](#)

Answer: c

Explanation: In the command the P gives condition to delete a particular tuple.

39. \_\_\_\_\_ resembles Create view.

- a) Create table . . . as
- b) Create view as
- c) Create table . . .like

d) With data

[View Answer](#)

Answer: a

Explanation: The 'create table... as' statement is similar to the 'create view... as' statement in that both are defined with queries. The main distinction is that table contents are set when the table is built, whereas view contents always reflect the current query result.

40. The query specifying the SQL view is said to be updatable if it meets which of the following conditions?

- a) select clause contains relation attribute names but not have expressions, aggregates, or distinct specification
- b) from clause has 1 relation
- c) query does not have group by or having clause
- d) All of the mentioned

[View Answer](#)

Answer: d

Explanation: To update the view in sql all the conditions must be satisfied.

41. When the "ROLLUP" operator for expression or columns within a "GROUP BY" clause is used?

- a) Find the groups that make up the subtotal in a row
- b) Create group-wise grand totals for the groups indicated in a GROUP BY clause
- c) Group expressions or columns specified in a GROUP BY clause in one direction, from right to left, for computing the subtotals
- d) To produce a cross-tabular report for computing subtotals by grouping phrases or columns given within a GROUP BY clause in all available directions

[View Answer](#)

Answer: c

Explanation: A view can be thought of as a virtual table that gets its data from one or more table columns.

42. Which of the following is the best way to represent the attributes in a large db?

- a) Dot representation
- b) Concatenation
- c) Relational-and
- d) All of the mentioned

[View Answer](#)

Answer: b

Explanation: Concatenation in DBMS is used to join two or more table fields of the same table or different tables. Example inst sec and student sec.

43. Which of the following is the subset of SQL commands used to manipulate Oracle Structures, including tables?

- a) Data Described Language
- b) Data Retrieval Language
- c) Data Manipulation Language
- d) Data Definition Language

[View Answer](#)

Answer: d

Explanation: DDLs are used to define schema and table characters and consist of commands that help in modifying. DML performs the change in the values of the relation.

44. Which of the following functions construct histograms and use buckets for ranking?

- a) Ntil()
- b) Newtil()
- c) Rank()
- d) All of the mentioned

[View Answer](#)

Answer: a

Explanation: ntile(n) returns the number of the bucket in which each tuple is stored, with bucket numbers beginning with 1.

45. \_\_\_\_\_ command is used in SQL to issue multiple CREATE TABLE, CREATE VIEW and GRANT statements in a single transaction.

- a) CREATE CLUSTER
- b) CREATE PACKAGE
- c) CREATE SCHEMA
- d) All of the mentioned

[View Answer](#)

Answer: c

Explanation: A schema is a description of a system's structure in a formal language supported by the database management system, and it refers to data organization as a blueprint for how a db is built.

46. Which of the following key is required in to handle the data when the encryption is applied to the data so that the unauthorised user cannot access the data?

- a) Primary key
- b) Authorised key
- c) Encryption key
- d) Decryption key

[View Answer](#)

Answer: d

Explanation: The encryption key is used to encrypt the message. Even if the message is intercepted by an enemy, the enemy will be unable to decrypt and interpret the message because he lacks the key.

47. Which of the following is known as the process of viewing cross-tab with a fixed value of one attribute?

- a) Dicing
- b) Pivoting
- c) Slicing
- d) Both Pivoting and Dicing

[View Answer](#)

Answer: c

Explanation: Slice procedure takes one dimension from a cube and turns it into a new sub-cube. Dice takes two or more dimensions from a cube and creates a new sub-cube from them.

48. For designing a normal RDBMS which of the following normal form is considered adequate?

- a) 4NF
- b) 3NF
- c) 2NF

d) 5NF

[View Answer](#)

Answer: b

Explanation: Because most 3NF tables are free of insertion, update, and deletion anomalies, an RDBMS table is sometimes regarded as “normalized” if it is in the Third Normal Form.

49. Which of the following is popular for applications such as storage of log files in a database management system since it offers the best write performance?

- a) RAID level 0
- b) RAID level 1
- c) RAID level 2
- d) RAID level 3

[View Answer](#)

Answer: b

Explanation: RAID level 0 refers to data striping. RAID level 1 refers to disk mirroring with block striping. RAID level 2 refers to bit-level striping and RAID level 3 refers to byte-level striping with dedicated parity.

50. Which of the following represents a query in the tuple relational calculus?

- a)  $\{ \{P(t) \mid t\} \}$
- b)  $\{t \mid P(t)\}$
- c)  $t \mid P() \mid t$
- d) All of the mentioned

[View Answer](#)

Answer: b

Explanation: A nonprocedural query language is the tuple relational calculus. It specifies the needed information but does not provide a detailed strategy for obtaining it.

51. The oldest DB model is \_\_\_\_\_

- a) Network
- b) Physical
- c) Hierarchical
- d) Relational

[View Answer](#)

Answer: a

Explanation: Network model has data stored in a hierarchical network flow. In a relational DBMS, the data is stored in the form of tables. Hierarchy is obtained by Parent-Child Relationship

52. Evaluate the statements issued by the DBA in the given sequence if OE and SCOTT are the users and the ORDERS table is owned by OE.

```
CREATE ROLE r1;  
GRANT SELECT, INSERT ON oe.orders TO r1;  
GRANT r1 TO scott;  
GRANT SELECT ON oe.orders TO scott;  
REVOKE SELECT ON oe.orders FROM scott;
```

What would be the outcome after executing the statements?

- a) The REVOKE statement would give an error because the SELECT privilege has been granted to the role R1
- b) The REVOKE statement would remove the SELECT privilege from SCOTT as well as

from the role R1

- c) SCOTT would be able to query the OE.ORDERS table
- d) SCOTT would not be able to query the OE.ORDERS table

[View Answer](#)

Answer: c

Explanation: To perform operations on objects, the REVOKE statement is used to revoke rights from a single user or role, or from all users.

53. Which of the following establishes a top-to-bottom relationship among the items?

- a) Relational schema
- b) Network schema
- c) Hierarchical schema
- d) All of the mentioned

[View Answer](#)

Answer: c

Explanation: A data model in which the data is structured into a tree-like structure is known as a hierarchical model. The structure allows information to be represented using parent-child relationships.

54. A major goal of the db system is to minimize the number of block transfers between the disk and memory. Which of the following helps in achieving this goal?

- a) Secondary storage
- b) Storage
- c) Catalog
- d) Buffer

[View Answer](#)

Answer: d

Explanation: Every block has a copy on disc, however, the copy on the disc may be an earlier version of the block than the version in the buffer.

55. What happens if a piece of data is stored in two places in the db?

- a) Storage space is wasted & Changing the data in one spot will cause data inconsistency
- b) It can be more easily accessed
- c) Changing the data in one spot will cause data inconsistency
- d) Storage space is wasted

[View Answer](#)

Answer: a

Explanation: One of the main features of a database management system is minimum data duplication and redundancy. Hence, it is always consistent and so there is no duplication.

56. The logical design, and the snapshot of the data at a given instant in time is known as?

- a) Instance & Relation
- b) Relation & Schema
- c) Domain & Schema
- d) Schema & Instance

[View Answer](#)

Answer: d

Explanation: Instance is an instance of time, the relation is also known as table consists

of data with similar characteristics, Domain is the collection of values that an attribute can contain and schema is a representation.

Ques No 1. The basic component of a file in a file system is a \_\_\_\_.

- (A) Data item
- (B) Field
- (C) Record
- (D) Tuple

**Show Answer**

**Answer: (A)**

Ques No 2. Database Management System is a \_\_\_\_ system to facilitate the creation and maintenance of a computerized database.

- (A) Physical
- (B) Software
- (C) Hardware
- (D) Knowledge

**Show Answer**

**Answer: (B)**

Ques No 3. UoD stands for \_\_\_\_.

- (A) Unordered Discourse
- (B) Union of Discourse
- (C) Universe of Discourse
- (D) None of the above

**Show Answer**

**Answer: (C)**

Ques No 4. The E-R data model is based on a perception of the real world that consists of a set of basic objects called \_\_\_\_.

- (A) Classes
- (B) Entities
- (C) Objects
- (D) Attributes

**Show Answer**

**Answer: (B)**

Ques No 5. The \_\_\_\_ level is at the highest level of database abstraction where only those portions of the database of concern to a user or application program are included.

- (A) External
- (B) Conceptual
- (C) Internal
- (D) Physical

**Show Answer**

**Answer: (A)**

Ques No 6. At \_\_\_\_ level of database abstraction, all the database entities and the relationships among them are included.

- (A) Physical
- (B) Conceptual
- (C) External
- (D) Internal

**Show Answer**

**Answer: (B)**

Ques No 7. In MySQL architecture, the function of the \_\_\_\_ is to keep copies of data for retrieval later, in case of a loss of data.

- (A) Transaction Manager
- (B) Buffer Manager
- (C) Recovery Manager
- (D) Query Engine

**Show Answer**

**Answer: (C)**

Ques No 8. \_\_\_\_ is a family of products that meet the data storage requirements of the largest data processing systems and commercial websites.

- (A) Microsoft Windows XP
- (B) Microsoft Windows ME
- (C) Microsoft Windows Vista
- (D) Microsoft SQL Server 2000

**Show Answer**

**Answer: (D)**

Ques No 9. In \_\_\_\_ data model, the objects are represented by record structures grouped in file structures.

- (A) Primitive
- (B) Classic
- (C) Semantic
- (D) Network

**Show Answer**

**Answer: (B)**

Ques No 10. A \_\_\_\_ is a collection of tuples, each of which contains values for a fixed number of attributes.

- (A) Association
- (B) Relation
- (C) Aggregation
- (D) None of the above

**Show Answer**

**Answer: (B)**

Ques No 11. \_\_\_\_ is a set of principles concerned with how data is structured.

- (A) Data integrity
- (B) Data manipulation
- (C) Data definition

(D) Data control

**Show Answer**

**Answer: (C)**

Ques No 12. \_\_\_\_ is a set of principles concerned with how data is operated upon.

(A) Data definition

(B) Data control

(C) Data integrity

(D) Data manipulation

**Show Answer**

**Answer: (D)**

Ques No 13. \_\_\_\_ is the fastest among the storage devices.

(A) Cache

(B) Main memory

(C) Flash memory

(D) Magnetic disk

**Show Answer**

**Answer: (A)**

Ques No 14. The contents of \_\_\_\_ are usually lost if a power failure or system crash occurs.

(A) Magnetic disk

(B) Main memory

(C) CD / DVD

(D) Tape storage

**Show Answer**

**Answer: (B)**

Ques No 15. \_\_\_\_ loses its contents when the power of the device is removed.

(A) Non-volatile storage

(B) Volatile storage

(C) CD

(D) Magnetic disk

**Show Answer**

**Answer: (B)**

Ques No 16. \_\_\_\_ retains its contents even when the power of the device is removed.

(A) Cache

(B) Volatile storage

(C) Non-volatile storage

(D) RAM

**Show Answer**

**Answer: (C)**

Ques No 17. A \_\_\_\_ is a collection of relations with distinct relation names.

(A) Domain

(B) Instance

(C) Schema

(D) Relational database

**Show Answer**

**Answer: (D)**

Ques No 18. The main construct for representing data in the relational model is a \_\_\_\_.

(A) Attribute

(B) Relation

(C) Tuple

(D) Domain

**Show Answer**

**Answer: (B)**

Ques No 19. A \_\_\_\_ is a column or a set of columns that refers to a primary key in the same table or another table.

(A) Primary key

(B) Foreign key

(C) Candidate key

(D) Concatenated key

**Show Answer**

**Answer: (B)**

Ques No 20. SQL is the \_\_\_\_ standard language for interacting with an RDBMS.

(A) OSI

(B) ISI

(C) ISO

(D) IEEE

**Show Answer**

**Answer: (C)**

Ques No 21. SQL stands for \_\_\_\_.

(A) Structured Query Language

(B) Standard Query Language

(C) Simple Query Language

(D) None of the above

**Show Answer**

**Answer: (A)**

Ques No 22. SQL commands can be roughly divided into \_\_\_\_ major categories concerning their functionality.

(A) Two

(B) Three

(C) Four

(D) Five

**Show Answer**

**Answer: (B)**

Ques No 23. \_\_\_\_ function is used to find the number of values in a column.

(A) SUM

(B) TOTAL

- (C) ADD
- (D) COUNT

**Show Answer**

**Answer: (D)**

Ques No 24. \_\_\_\_ function is used to find the total of all the values in a column.

- (A) SUM
- (B) COUNT
- (C) ADD
- (D) AVG

**Show Answer**

**Answer: (A)**

Ques No 25. Union operation is represented by the \_\_\_\_ symbol.

- (A) U
- (B)  $\mu$
- (C)  $\Omega$
- (D)  $\varepsilon$

**Show Answer**

**Answer: (A)**

Ques No 26. The \_\_\_\_ operation removes common tuples from the first relation.

- (A) Union
- (B) Difference
- (C) Cartesian Product
- (D) Projection

**Show Answer**

**Answer: (B)**

Ques No 27. TRC stands for \_\_\_\_.

- (A) Tuple Relation Calculation
- (B) Triple Relational Calculus
- (C) Tuple Relational Calculus
- (D) Triple Relation Calculation

**Show Answer**

**Answer: (C)**

Ques No 28. In the atomic formula clauses, the quantifiers “For any” and “For all” are said to \_\_\_\_ the variable R.

- (A) Destroy
- (B) Bind
- (C) Separate
- (D) None of the above

**Show Answer**

**Answer: (B)**

Ques No 29. Y is said to be functionally dependent on X if \_\_\_\_.

- (A) Y has only one value corresponding to the value of X
- (B) Y has two values corresponding to the value of X
- (C) Y has three values corresponding to the value of X

(D) Y has four values corresponding to the value of X

**Show Answer**

**Answer: (A)**

Ques No 30. X, Z\_Y indicates that functional dependency may also be based on a composite attribute. State True or False.

(A) False

(B) True

(C) Either (A) or (B)

(D) Unpredictable

**Show Answer**

**Answer: (B)**

Ques No 31. Given a relation, if the value of an attribute X uniquely determines the value of all other attributes in a row, then X is said to be \_\_\_\_.

(A) Primary key

(B) Foreign key

(C) Relation key

(D) Key

**Show Answer**

**Answer: (C)**

Ques No 32. Let X and Y be the two attributes of a relation. The functional dependency can be written as \_\_\_\_.

(A) X\_Y

(B) X\_\_Y

(C) X==Y

(D) X\_Y

**Show Answer**

**Answer: (D)**

Ques No 33. A distributed database system consists of a collection of sites, each of which maintains a \_\_\_\_ database system.

(A) Local

(B) Remote

(C) Administrator

(D) None of the above

**Show Answer**

**Answer: (A)**

Ques No 34. The site of a distributed database system that may be distributed physically over a large geographical area is called \_\_\_\_.

(A) Short-haul network

(B) Long-haul network

(C) Medium-haul network

(D) Large-haul network

**Show Answer**

**Answer: (B)**

Ques No 35. The frequency with which a link or site fails is measured by \_\_\_\_.

(A) Availability

- (B) Efficiency
- (C) Reliability
- (D) Affectivity

**Show Answer**

**Answer: (C)**

Ques No 36. The cost of physically linking the sites in the system is \_\_\_\_.

- (A) Hardware cost
- (B) Software cost
- (C) Communication cost
- (D) Installation cost

**Show Answer**

**Answer: (D)**

Ques No 37. CAD is an example for \_\_\_\_.

- (A) Design databases
- (B) Multimedia databases
- (C) Knowledge databases
- (D) None of the above

**Show Answer**

**Answer: (A)**

Ques No 38. Artificial intelligence and expert systems fall under \_\_\_\_.

- (A) Design databases
- (B) Knowledge databases
- (C) Multimedia databases
- (D) None of the above

**Show Answer**

**Answer: (B)**

Ques No 39. Pick out the object-oriented programming language.

- (A) BASIC
- (B) FORTRAN
- (C) C++
- (D) COBOL

**Show Answer**

**Answer: (C)**

Ques No 40. The combination of attributes and methods is called \_\_\_\_.

- (A) Class
- (B) Entity
- (C) Software
- (D) Object

**Show Answer**

**Answer: (D)**

Ques No 41. \_\_\_\_ were among the first to use databases in a geographically distributed manner.

- (A) Railways
- (B) Waterways
- (C) Airlines

(D) Roadways

**Show Answer**

**Answer: (C)**

Ques No 42. \_\_\_\_ data independence implies that application programs need not be changed if fields are added to an existing record.

(A) Physical

(B) Static

(C) Dynamic

(D) Logical

**Show Answer**

**Answer: (D)**

Ques No 43. \_\_\_\_ level is at the lowest level of abstraction, closest to the physical storage method used.

(A) Physical

(B) Conceptual

(C) Internal

(D) External

**Show Answer**

**Answer: (C)**

Ques No 44. In MySQL architecture \_\_\_\_ interfaces with the operating system to write data to the disk efficiently.

(A) Query Engine

(B) Buffer Manager

(C) Transaction Manager

(D) Storage Manager

**Show Answer**

**Answer: (D)**

Ques No 45. \_\_\_\_ model organizes data in a tree structure.

(A) Hierarchical

(B) Object/Relational

(C) Network

(D) Object-Oriented Model

**Show Answer**

**Answer: (A)**

Ques No 46. \_\_\_\_ model was formally defined by Conference on Data Systems Languages (CODASYL) in 1971.

(A) Hierarchical

(B) Object/Relational

(C) Object-Oriented

(D) Network

**Show Answer**

**Answer: (D)**

Ques No 47. \_\_\_\_ is also called Electrically Erasable Programmable Read-Only Memory (EEPROM).

(A) Main memory

- (B) Cache
- (C) Flash memory
- (D) Magnetic disk

**Show Answer**

**Answer: (C)**

Ques No 48. \_\_\_\_ is referred to as sequential access storage.

- (A) Optical storage
- (B) Magnetic disk storage
- (C) Cache
- (D) Tape storage

**Show Answer**

**Answer: (D)**

Ques No 49. An instance of a relation is a set of \_\_\_\_.

- (A) Fields
- (B) Tuples
- (C) Columns
- (D) Values

**Show Answer**

**Answer: (B)**

Ques No 50. Tuples can also be called \_\_\_\_.

- (A) Fields
- (B) Values
- (C) Columns
- (D) Records

**Show Answer**

**Answer: (D)**